FAST Direct Hot-Pressing brings Sintering up to Speed

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The unique FAST-sintering process is employed in an increasing number of industries and fields of R&D. Up to now, almost 1.000 sinter presses have been installed worldwide and new applications are added constantly.

Having its origin in the diamond tool industry, FAST Direct Hot-Pressing has spread into many other industries. Back in 1953, not even the inventors would have imagined the diverse fields of application for this sintering technique. Today, this technology is entering such diverse fields as the diamond tools and friction materials industry, sputter target manufacturing, heat sink production, the hard metal and ceramics industry as well as universities and R&D-institutes. The list is getting longer every year and with new developments coming up it will probably never be completed.

Index Terms* — *field assisted sintering – hot pressing – production technology – alternative to spark plasma sintering

I. Introduction

Most users are already familiar with traditional hot-pressing techniques before switching to FAST Direct-Hot-Pressing. Hot-pressing is one of the widely established methods used to consolidate cold-pressed or loose powders.
However, most people associate hot-pressing with long sinter cycles and low flexibility. Normally, large furnaces are heated up via induction coils or graphite heating elements from outside to inside. First, the walls of the chamber are heated up, then the atmosphere inside the chamber, then the mold and in the very end the powder or green compact inside the mold. It is obvious that this sinter cycle takes comparably long time and changing the temperature during or between the cycles is a slow and inaccurate process due to the sluggish nature of the system.

FAST Direct Hot-Pressing takes another approach. FAST stands for "Field Assisted Sintering Technique". The mold is directly connected to electric power. The resistivity of the mold material and the powder part or green compact generate the heat directly inside the mold. Not the furnace, nor the atmosphere inside the furnace are heated up. The heat is only generated where it is needed.
This results in very high heating rates and additionally a significant increase in the sintering activity of fine metal powder aggregates. Depending on the part size, sinter cycles of only a few minutes can be achieved. Further, this process lowers the sintering temperature and pressure compared to that required in traditional sintering processes.

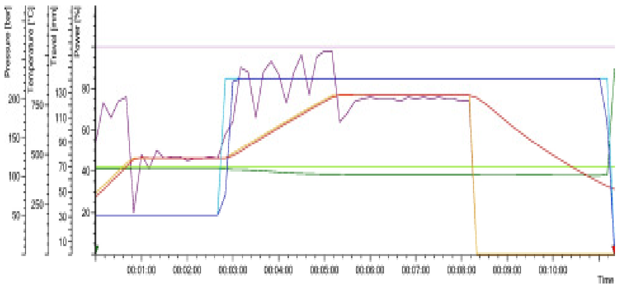

Fig. 1: Typical FAST-DHP sinter cycle of a Co-bond (50mm diameter) sintered in 11,5 minutes.

II. FIELD ASSISTED SINTERING TECHNIQUE

The basic idea of sintering with electric current is quite old. Resistance heating of hard metal powders was patented by American inventor George F. Taylor as early as 1933[1]. Since then, the technique evolved and several variations of Taylor`s idea are available nowadays. When applying a standard (unpulsed) current, the technique is often referred to as "FAST Direct Hot-Pressing (FAST DHP)". Another common term is "Rapid Hot-Pressing (RHP)". When applying a pulsed current, it is referred to as "Spark Plasma Sintering (SPS)". Due to the high cost related to generating a pulsed current and the unproven effect of it, this technique is less common for industrial applications.
All these variations of Direct Hot-Pressing are summarized under the generic term "Field Assisted Sintering Technique (FAST)"[2].
Latest research suggests that there is no evident difference between sintering with pulsed or unpulsed current (SPS or FAST DHP). The same improved sinter results (compared to traditional sintering) can be achieved by all FAST sintering techniques [3].

[1] G.F. Taylor, U.S. patent 1896854, "Apparatus for Making Hard Metal Compositions", February 7th, 1933
[2] Ref. to Wikipedia, Article on „Hot Pressing", as of August 12th, 2013
[3] „International Powder Metallurgy Directory" (January 4th, 2012): 2011 Hagen Symposium. A Review of Spark Plasma Sintering by Prof. Bernd Kieback, Director of Fraunhofer IFAM branch lab Dresden and the Institute for Materials Science at the Technical University of Dresden (Germany). Summary has been published by Dr. Georg Schlieper.

The temperature range of FAST Direct Hot-Pressing is determined by the pyrometer and construction of the sinter chamber. Usually, the machines are available in two versions: a low sinter temperature version for up to 1.100°C (~2.000°F) and a high sinter temperature version for up to 2.400°C (~4.350°F). The difference is the construction of the sinter chamber, especially the heat insulation, and the temperature measurement devices (thermocouples or pyrometer). FAST sintering presses normally apply a uniaxial pressing force with an hydraulic cylinder moving the upper or –less common- the lower ram of the machine. Standard pressing forces can be as high as 200t. One needs to keep in mind that FAST Direct Hot-Pressing allows to sinter most materials to full density at lower temperature and lower pressure compared to traditional hot-pressing techniques. Therefore, the high pressing forces of typical conventional presses are not required.

The mold material is normally graphite, a material which is available worldwide at low cost in standard qualities. Using metal powders, the conductivity of the mold is ideal for FAST-sintering of the workpiece. Molds with a big diameter and even with several layers can be heated up very quickly. The process is especially suitable for materials which require high heating rates, e.g. for materials that should not be kept at high temperatures. For metal powder workpieces, a typical sinter cycle takes about 10-15 minutes for small geometries (e.g. 100x100mm). For larger geometries like diameter 200mm, a typical cycle takes approximately 45 minutes. Sintering several layers simultaneously does not necessarily prolong the cycle time dramatically. In fact, sintering more than one layer can even have a positive effect on the power consumption because working with stacks increases the electric resistivity.

Typical materials and applications of
FAST Direct Hot-Pressing are:

- Hard metals and steels
- Multi-layer composite materials
- Functionally Graded Materials (FGMs)
- High-performance ceramic materials
- Special materials to be sintered with minor grain growth
- Sputter targets
- Friction materials for brakes and clutches
- Diamond tools (metal-diamond-alloys)
- Heat sinks (copper-diamond-alloys)

The short sinter cycles of the FAST-sintering process make the technique ideally suitable for sintering all kind of metal powders. No wonder that this is one of the main applications. The short cycles reduce grain growth to an absolute minimum. This ensures that the material keeps the desired characteristics and helps to achieve the required density.
Very often, binderless sintering is possible, so the use of Nickel and Cobalt can be avoided.

Due to the good electrical conductivity of the material, all advantages of a FAST Direct Hot-Pressing process are coming into effect, mainly:

- Reduction of grain growth thanks to steep heating rates
- Effective use of energy because heat is generated where it is needed.

When talking metals or metal alloys, most materials are tungsten carbide, pure tungsten, tungsten-titanium or (stainless) steels. With all these materials excellent sinter results can be achieved. In some industries, a good result is defined as the full densification of the material, e.g. in the sputter target industry. For other applications, the material must have an exactly defined porosity, for example for stainless steel filters. The exact control of all process parameters is a very important reason why researchers as well as manufacturers switch from un-precise traditional furnaces to modern-style FAST Direct Hot-Presses.

Besides (hard) metals and steel, typical applications include the sintering of technical ceramics, such as B_4C, AlN, TiB_2, Al_2O_3 and others. For sputter target applications, AlN has already been sintered with 300mm diameter using a Dr. Fritsch FAST Direct Hot-Press which proves the technology not only suitable for sintering large dimensions in metals, but also in ceramic materials.

Some up-to-date fields of research which require the use of FAST sintering machines are so called Functionally Graded Materials (FGMs). A product may have the characteristics and advantages of material A on one side and of material B on the other side. Between these two sides, there is a material gradient.
Another example are thermo-electric materials, which create electricity when being heated up.
In both cases, the sintering process must be fast enough to avoid undesired grain growth. This is achieved by FAST Direct Hot-Pressing. This technique creates a controlled liquid phase between the different materials, allowing them to bond together. Furthermore, the layers of consolidated loose powder can be sintered onto a carrier.

III. CONCLUSION

FAST Direct Hot-Pressing is the answer to many issues industries and researchers around the world are dealing with. It allows to sinter materials with exactly pre-defined characteristics and combines this precision with a very fast and energy-efficient manufacturing process. The development of new materials can be accelerated significantly compared to traditional hot-pressing techniques. In fact, the sinter quality and achievable density comes close or is comparable to Hot-Isostatic-Pressing (HIP) but at a fraction of the cost and with much less grain growth.